# Architecture Style Retrieval System

# 🛠️ How to Use It

## ⚠️If you download it and run it directly, make sure it's on macOS!

## We haven't tested if it works on windows, you need to replace all the paths with ones that work on windows.

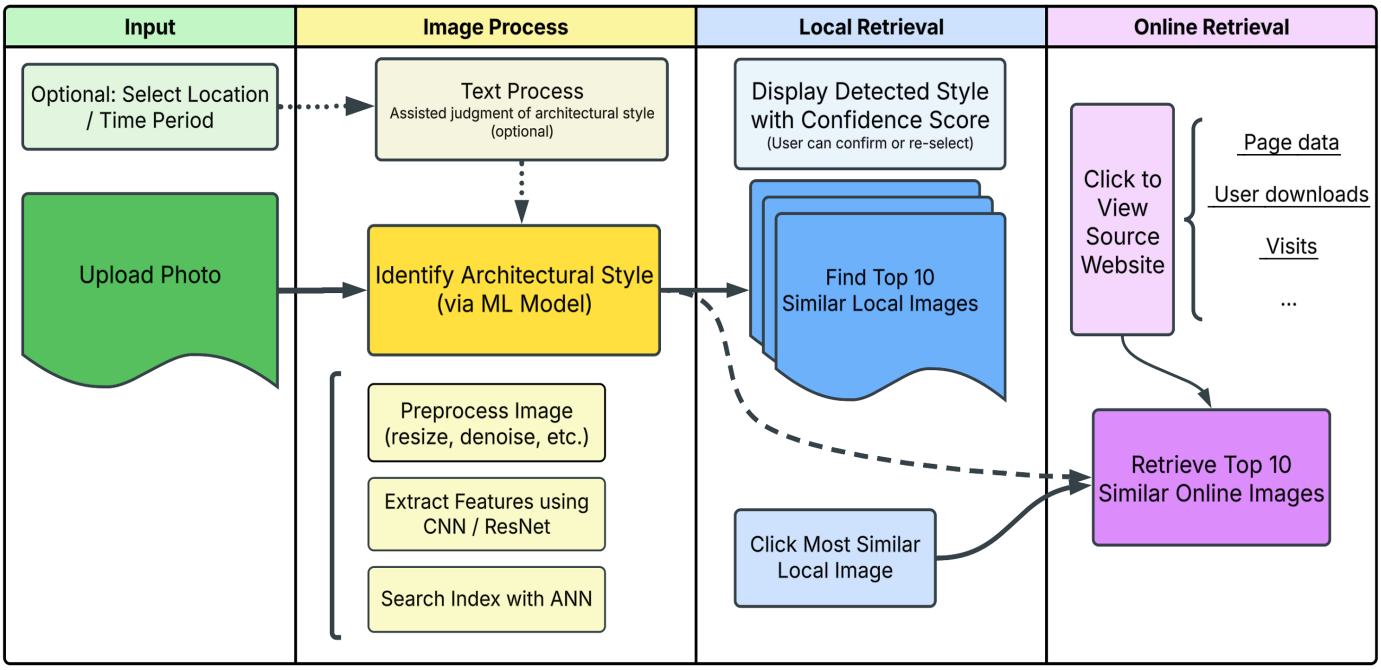
1. **Upload an Image**
   Drag and drop or select a building image to upload via the main interface. Once uploaded, the system will automatically classify the architectural style of the image and display the prediction along with a confidence score.
2. **View Top-10 Local Similar Images**
   In the **Top 10 Local Similar Images** section, you can explore visually similar images retrieved from the local dataset. Each image includes its predicted style, similarity score, and a download link. You can click the ⭐ **Favorite** button to select the one you find most similar.
3. **Retrieve Top-K Similar Online Images**
   In the **Top K Similar Online Images** section, the system performs an online search based on your favorited image. You can also enhance the search by entering additional textual hints in the sidebar, such as:
   - **Location / Style** (e.g., "France", "Gothic")
   - **Time Period** (e.g., "19th century", "night view")
   - **Top K Results** (1–10)

   The retrieved images will include direct source links for further reference.

# Catalogs

```
├─ dataset
│   └─ data_clean
│      └─ rename_images.py
│
├─ images
│   ├─ bgimage
│   ├─ readme_img
│   ├─ test_image
│   └─ usydlogo.png
│
├─ MR                          # main project
│   ├─ main.py                   # entrances
│   └─ ui_v4.py                  # Web UI
│
├─ part1_model                           # Model Training and Feature Extraction Module
│   ├─ best_grid_50vit_deep.pth   # model weight
│   ├─ best_hyperparams_deep.json     # best parameter
│   ├─ feature50.py
│   ├─ feature_vit50_new.py
│   ├─ features_50vit.csv        # Extracted features (you can ignore the upload)
│   ├─ trainer.py
│   └─ trainer_deeper.py
│
├─ part2_localsearch             # Local Image Retrieval Module
│   ├─ local_search.py
│   └─ local_top.py
│
├─ part3_onlinesearch            # Online image retrieval module (multimodal)
│   └─ online_multi_mod.py
│
├─ README.md
└─ requirements.txt
```

# Flow Chart


Input
Image Process
Local Retrieval
Online Retrieval
Optional: Select Location / Time Period
Text Process
Assisted judgment of architectural style (optional)
Display Detected Style with Confidence Score
(User can confirm or re-select)
Click to View Source Website
Page data
User downloads
Visits
...
Upload Photo
Identify Architectural Style (via ML Model)
Find Top 10 Similar Local Images
Preprocess Image (resize, denoise, etc.)
Extract Features using CNN / ResNet
Search Index with ANN
Retrieve Top 10 Similar Online Images
Click Most Similar Local Image

# Setup

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### You need to wait for your environment to install the python packages (everything needed is already listed in requirements.txt)

### When everything is ready, run the main.py file and wait for the default browser to pop up the WebUI.

### (Everything works as in the video.)